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Local three-dimensional earthquake tomography by trans-dimensional Monte Carlo sampling

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SUMMARY

Local earthquake tomography is a non-linear and non-unique inverse problem that uses event arrival times to solve for the spatial distribution of elastic properties. The typical approach is to apply iterative linearization and derive a preferred solution, but such solutions are biased by a number of subjective choices: the starting model that is iteratively adjusted, the degree of regularization used to obtain a smooth solution, and the assumed noise level in the arrival time data. These subjective choices also affect the estimation of the uncertainties in the inverted parameters. The method presented here is developed in a Bayesian framework where *a priori* information and measurements are combined to define a posterior probability density of the parameters of interest: elastic properties in a subsurface 3-D model, hypocentre coordinates and noise level in the data. We apply a trans-dimensional Markov chain Monte Carlo algorithm that asymptotically samples the posterior distribution of the investigated parameters. This approach allows us to overcome the issues raised above. First, starting a number of sampling chains from random samples of the prior probability distribution lessens the dependence of the solution from the starting point. Secondly, the number of elastic parameters in the 3-D subsurface model is one of the unknowns in the inversion, and the parsimony of Bayesian inference ensures that the degree of detail in the solution is controlled by the information in the data, given realistic assumptions for the error statistics. Finally, the noise level in the data, which controls the uncertainties of the solution, is also one of the inverted parameters, providing a first-order estimate of the data errors. We apply our method to both synthetic and field arrival time data. The synthetic data inversion successfully recovers velocity anomalies, hypocentre coordinates and the level of noise in the data. The Bayesian inversion of field measurements gives results comparable to those obtained independently by linearized inversion, reconstructing the geometry of the main seismic velocity anomalies. The quantification of the posterior uncertainties, a crucial output of Bayesian inversion, allows for visualizing regions where elastic properties are closely constrained by the data and is used here to directly compare our results to the ones obtained with the linearized inversion. In the case we examined the results of two inversion techniques are not significantly different.

Key words: Seismic tomography; Computational seismology; Crustal structure.

1 INTRODUCTION

Local earthquake tomography (LET) is widely used to image subsurface elastic properties using natural seismicity. In the most common application, arrival times of *P* waves in a network of seismometers are used to map positive and negative velocity anomalies with respect to a reference velocity model (Kissling 1988). The subsurface volume is divided into cells and the arrival time of a seismic wave is dependent on the seismic velocities of all the cells crossed by the ray. Many different seismic rays crossing the crustal volume with different trajectories are needed to constrain seismic velocity anomalies at depth and reduce the non-uniqueness of the problem (Thurber 1993). This tomographic inversion method has been successfully applied from the late 1970s, when facilities for intensive computation became widely available. The solution of the LET inverse problem needs to deal with two fundamental issues. First, the position of the sources is an unknown itself, which must be determined within the inversion process. This results in a non-unique inverse problem (Eberhart-Phillips 1990) and increases the uncertainties associated with the retrieved velocity models. Secondly, the

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geometry of a seismic ray, and hence its arrival time, is a non-linear function of the velocity of the media in which it propagates (Thurber 1993). Classical approaches to LET are based on a linearization of the tomographic inverse problem and use a smoothing regularization to stabilize the solution. Final results are usually given in the form of perturbations of a background elastic model by trading off smoothness and fit to the data. In the last decades, modifications to this approach have been applied to tackle the non-linearity and non-uniqueness of the seismic tomography inverse problem, using, for example an adaptive-grid formalism (Michelini 1995), deformable layer tomography (Zhou *et al.* 2010), or Voronoi tessellation (Sambridge & Gudmundsson 1998; Sambridge & Faletic 2003; Zhang & Thurber 2005; Bodin *et al.* 2009). Although these approaches, based on an irregular parametrization of the Earth volume, help to reduce the dimensionality of the parameter space, damping or smoothing factors are still required to obtain stable solutions. Determining the values of such factors requires great care and it is often speculative.

We develop here a stochastic algorithm for a fully non-linear 3-D LET via trans-dimensional Monte Carlo sampling of the subsurface elastic properties and of the seismic event locations. We adopt the framework of Bayesian inference, where information on the parameters of interest is quantified by their probability distributions. We describe and implement a reversible jump Markov chain Monte Carlo (MCMC) algorithm where the dimensionality of the earth models can vary during the sampling (Geyer & Møller 1994; Green 1995). Such choice results in an adaptive parametrization of the elastic properties so that the resolution of the velocity model is controlled by the data (Bodin *et al.* 2009). As the noise level in the arrival time data is typically poorly known *a priori*, the algorithm applies a hierarchical Bayes approach, where the noise level is treated as an unknown itself (e.g. Malinverno & Briggs 2004; Dettmer *et al.* 2007; Bodin *et al.* 2012; Dettmer *et al.* 2012). After describing the algorithm, we apply it to synthetic data and field measurements that were previously inverted with a linearized approach.

1.1 Background: Bayesian approach, trans-dimensional MCMC and seismic tomography

Monte Carlo sampling methods are an alternative to linearization and are usually developed to solve inverse problems from a Bayesian point of view, where the focus is the full posterior probability density (PPD) of the parameters of interest rather than a single optimal value (e.g. Tarantola 2006). Monte Carlo sampling has been widely applied to solve non-linear and non-unique inverse problems in geophysics (e.g. Press 1968; Khan *et al.* 2000; Spada 2001; Sambridge & Mosegaard 2002). MCMC algorithms follow a random walk in model space according to probabilistic rules designed to produce a chain or sequence of sampled parameters that are asymptotically distributed as in the PPD (Mosegaard & Tarantola 1995). Sampling requires solving the forward problem at each step, and the computational cost of the forward prediction is the main drawback of the Monte Carlo approach. However, such computations can be implemented on a cluster, and present-day facilities for parallel computing can alleviate this disadvantage.

In recent years, reversible jump Markov chain Monte Carlo (RJMCMC) algorithms (Geyer & Møller 1994; Green 1995) have been introduced to address the trans-dimensional case where the number of parameters in the inverse problem is itself unknown. Trans-dimensional RJMCMC algorithms have been applied widely in geophysical inversion (e.g. Malinverno 2002; Malinverno & Leaney 2005; Sambridge *et al.* 2006; Hopcroft *et al.* 2007; Piana Agostinetti & Malinverno 2010; Gallagher 2012; Ray & Key 2012; Galletti *et al.* 2013; Sambridge *et al.* 2013). The attractive feature of a trans-dimensional approach is that the level of detail in the solution is locally controlled by the data and not imposed by a global, subjective degree of smoothing. However, the complexity of the solution is tightly linked to the assumed uncertainties in the data (e.g. Dettmer & Dosso 2012), which are typically not well known *a priori*. To address this issue, a hierarchical Bayes inversion scheme has been introduced where parameters characterizing the data uncertainties (e.g. the standard deviation of the data errors) are treated as unknowns, and estimates of these parameters can be inferred from MCMC sampling (e.g. Malinverno & Briggs 2004; Bodin *et al.* 2012).

Monte Carlo methods have been applied to seismic tomography both on local and regional scales. For example, Debski (2010) used a MC algorithm to invert data from microseismicity and obtain a map of seismic velocity in a mine. An adaptive MC sampling scheme has been adopted to explore 2-D crustal velocity structure by joint refraction and reflection seismic tomography in Korenaga & Sager (2012). Seismic tomography using a RJMCMC algorithm has been developed in Bodin *et al.* (2009, 2012) to invert data from ambient noise analysis. To our knowledge, the study of Bodin *et al.* (2009) has been the first attempt to apply a trans-dimensional algorithm to a 2-D geophysical inverse problem. In Bodin *et al.* (2012), multiscale inversion of different data sets was achieved with a hierarchical Bayes approach and the RJMCMC algorithm was applied in an iterative scheme where ray paths were updated at each iteration but fixed during the RJMCMC sampling. The method described here will also apply a hierarchical Bayes approach to determine the noise level in the data and will use a similar updating scheme for the ray paths. Our main new contribution is the application of RJMCMC to the simultaneous inversion of a fully 3-D velocity structure and event locations, that is where the model parameters are not only related to the elastic model, but also to the position of the seismic sources.

2 METHOD

In this section, we illustrate the details of our new algorithm within the framework of Bayesian inference. The algorithm applies a classical MCMC sampling scheme adapted to address the LET inverse problem. We first introduce the concept of Voronoi tessellation, a useful tool for adaptive gridding of a 3-D volume (e.g. Sambridge *et al.* 1995), and define our subsurface model as an ensemble of Voronoi cells. We then illustrate the forward process, that is how we compute the predicted *P*-wave arrival times for a given subsurface model. Finally, we describe our recipe for trans-dimensional, RJMCMC sampling and briefly discuss the convergence issue, that is how long the sampling should be run, and hence the computational cost of the algorithm.

2.1 Model parametrization

The LET inverse problem is non-unique due to the trade-off between the position of the seismic source and the seismic velocity along the ray path between the event and the stations. We invert for both the subsurface elastic properties and the location of the seismic events, accounting for the dependence of the ray paths from both sets of parameters. The trade-off between parameters can be investigated from the analysis of the PPD.

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Solving the LET inverse problem with a RJMCMC algorithm requires an adaptive ensemble of elastic nuclei that define the 3-D distribution of elastic properties in the investigated volume. Both the nuclei distribution and their number can change during sampling, so that the nuclei density in the volume is dictated by the data, reaching high resolution where needed or defining a sparser mesh where few data are available (Bodin *et al.* 2012). We use Voronoi tessellation, which is a natural way to completely cover a volume with non-overlapping, convex cells from a distribution of points in 3-D space (e.g. Sambridge *et al.* 1995).

In our algorithm, the model vector $\mathbf{m}$ consists of two scalars and three vectors as in

$$\mathbf{m} = (\sigma_{\text{err}}, k, \mathbf{s}, \mathbf{v}, \mathbf{e}). \tag{1}$$

The scalar σ_{err} is the standard deviation of the residual errors, defined as the differences between measured and predicted *P*-wave arrival times. The scalar k is the number of elastic nuclei in the investigated volume. The position of the elastic nuclei is given by the $3k$-vector $\mathbf{s} = (x_1, y_1, z_1, \ldots, x_k, y_k, z_k)$. The seismic velocities are in the k-vector $\mathbf{v} = (V_1, \ldots, V_k)$, which contains the values of *P*-wave velocity within each Voronoi cell. Finally, the vector $\mathbf{e} = (e_{x,1}, e_{y,1}, e_{z,1}, \ldots, e_{x,N_e}, e_{y,N_e}, e_{z,N_e})$ contains the coordinates of the N_e events.

2.2 Measurements and forward modelling

The key measurements are *P*-wave arrival times on a network of seismic stations, collated in a data vector $\mathbf{d}$ of N_d event-station pairs. To evaluate how a given velocity model $\mathbf{m}$ fits the data, these observed arrival times are compared to predicted arrival times. The predicted *P*-wave arrival time $g_i(\mathbf{m})$ for the ith event-station pair equals the corresponding event origin time plus the traveltime through $\mathbf{m}$ computed by ray tracing. Our algorithm differs from that of Bodin *et al.* (2012), who compute predicted ray paths in a reference velocity model. We adopted a more conservative approach, computing the ray paths in all the candidate models along the MCMC sampling. This choice increases the run-time of the inversion, but prevents the introduction of possible bias from the approximation implicit in the approach of Bodin *et al.* (2012).

For the origin time, we adopt the 'minimum origin time' criterion (Billings *et al.* 1994), where the origin time of an event is determined for a given velocity model by maximizing the likelihood (see Section 2.3), so that the origin time is not one of the inverted parameters. It has been shown that such choice helps the convergence and allows for escaping from local minima given by the strong trade-off between velocity and event locations (Giacomuzzi 2013).

For ray tracing, we use the ray tracer implemented for the code *SimulPs12* (Evans *et al.* 1994). The *P*-wave velocity field in the subsurface, represented by the ensemble of Voronoi cells, is rasterized over a Cartesian 3-D grid of nodes where the *P*-wave velocity value is the value at the nearest Voronoi nucleus. The 3-D grid spacing is set to be less than the minimum distance between two nuclei in the Voronoi cells, ensuring that each Voronoi cell is mapped by at least one grid node. The grid spacing should match the highest spatial resolution expected in the elastic parameters, given the distribution of the events and of the seismic stations. To account for the higher resolution expected in the central part of our investigated volumes, our 3-D grid is more densely spaced at the centre of the domain. In Fig. 1, we plot one of the models sampled by our algorithm on vertical and horizontal sections. Colours refer to V_{P} values inside the Voronoi cells and white crosses mark the nodes in the 3-D grid. All Voronoi cells are sampled by the 3-D grid and thus contribute to the computation of the synthetic data (i.e. to the likelihood of the model). As noted by Bodin *et al.* (2009), any sampled Voronoi model does not allow any meaningful inference on a parameter of interest, and only the global ensemble (all the models for all the chains) should be considered. The 3-D grid should be fine enough to represent the information contained in the data, rather than to reproduce in detail the geometry of the Voronoi cells. Fine details of the Voronoi cells are usually not supported by realistic data (and errors) and disappear in the final ensemble which represents the PPD. Moreover, in the *SimulPs12* ray tracer the 3-D grid is resampled at a fixed interval set here at 1.0 km and velocity is linearly interpolated within the resampled grid. This fine discretization ensures that perturbations of the elastic model are coupled to perturbations of the event positions. Pseudo-bending of the seismic ray is obtained by an iterative minimization of the traveltime.

2.3 Bayesian approach, RJMCMC and hierarchical Bayes

As the LET inverse problem is intrinsically non-unique (e.g. Kissling 1988), optimization algorithms that focus on a single solution cannot explore the trade-off between investigated parameters. A probabilistic approach, where knowledge of the investigated parameters is expressed in terms of probability distributions, is a more informative way to solve such an inverse problem. The aim is to understand how (and if) the data sharpen our prior knowledge about the parameters of interest. The PPD is given by Bayes's rule as

$$p(\mathbf{m}|\mathbf{d}, \mathcal{I}) = \frac{p(\mathbf{m}|\mathcal{I})p(\mathbf{d}|\mathbf{m}, \mathcal{I})}{p(\mathbf{d}|\mathcal{I})} \tag{2}$$

where $p(\,)$ is a probability density, $A|B$ indicates conditional dependence (i.e. A given B), $\mathbf{d}$ denotes the observed data and $\mathcal{I}$ the prior information. Thus, $p(\mathbf{m}|\mathbf{d}, \mathcal{I})$ is the target PPD, $p(\mathbf{m}|\mathcal{I})$ represents the prior information available on the model and $p(\mathbf{d}|\mathbf{m}, \mathcal{I})$ is the likelihood, which is the probability to obtain the observed data $\mathbf{d}$ for a given model $\mathbf{m}$. The denominator $p(\mathbf{d}|\mathcal{I})$ is called 'evidence' or marginal likelihood. It is a normalization factor that makes the integral of the PPD over the model space equal to unity and in practice it is often difficult to evaluate. Although evidence is usually important in model selection, MCMC methods are based on the ratio of the PPDs of different models and do not need to compute the evidence integral.

Posterior information on the parameters is usually extracted from summary properties of the PPD, such as marginal probability density function (PDF) and common estimators such as posterior means and standard deviations. As the frequency distribution of the parameter values sampled by MCMC approximates the PPD, these summary properties are easy to compute: the marginal PPD of a parameter is approximated by its sample histogram, and the posterior mean and the standard deviation by the corresponding sample values.

Our algorithm is based on the Metropolis-Hastings algorithm (Metropolis *et al.* 1953; Hastings 1970; Chib & Greenberg 1995). Briefly, starting from the current model $\mathbf{m}_{\text{curr}}$, a new 'candidate' model $\mathbf{m}_{\text{cand}}$ is proposed that is a perturbation of the current model. The candidate model is accepted and the chain samples the new point with an 'acceptance probability' α that is equal to the ratio between the product of the proposal function and the PPD, for the candidate and for the current model (Green 1995). If the candidate model is rejected, the chain resamples the current model, a new candidate is proposed, and so on. We follow the MCMC sampling strategy given by Mosegaard & Tarantola (1995), where the

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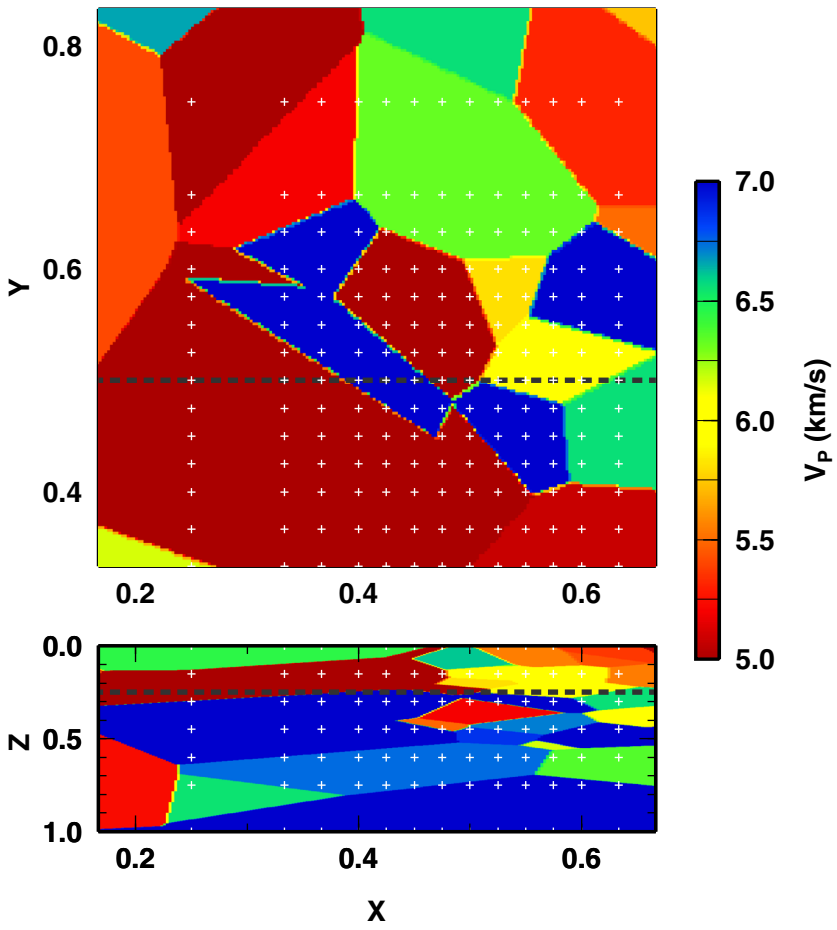


Figure 1. Voronoi diagram for a model sampled along the RJMCMC. (a) Map view of the seismic velocity at 2.5 km depth. Colours represent *P*-wave velocity values. Dashed grey line indicates the position of the vertical section in (b). White crosses show the *XY* position of the grid node. (b) Vertical section of the seismic velocity along the dashed line in (a). Dashed grey line shows the position of the map in (a). Distance is expressed in a-dimensional units, while the ratio between axis lengths are maintained (i.e. *Z*-axis is shorter).

process of generating candidate models is designed to sample their prior distribution. In this case, the acceptance probability α for the candidate model is simply the ratio between the likelihoods of the candidate and of the current model.

This MCMC algorithm can be extended to sample a parameter space where the models can have a different number of dimensions. Green (1995) gave the theoretical basis for 'reversible jump' MCMC (RJMCMC), where the dimension of the models can also be perturbed when proposing candidates. The general formula for the acceptance probability in RJMCMC is (Green 1995; Gallagher *et al.* 2009)

$$\alpha = \min\left[1, \frac{q(\mathbf{m}_{\rm curr}|\mathbf{m}_{\rm cand})}{q(\mathbf{m}_{\rm cand}|\mathbf{m}_{\rm curr})} \frac{p(\mathbf{m}_{\rm cand}|\mathcal{I})}{p(\mathbf{m}_{\rm curr}|\mathcal{I})} \frac{p(\mathbf{d}|\mathbf{m}_{\rm cand},\mathcal{I})}{p(\mathbf{d}|\mathbf{m}_{\rm curr},\mathcal{I})} |\mathbf{J}|\right], \quad (3)$$

where $p(\mathbf{m}|\mathcal{I})$ is the prior distribution, $p(\mathbf{d}|\mathbf{m},\mathcal{I})$ is the likelihood, $q(\mathbf{m}_{\rm cand}|\mathbf{m}_{\rm curr})$ is the proposal distribution (the probability of obtaining a candidate $\mathbf{m}_{\rm cand}$ when the current model is $\mathbf{m}_{\rm curr}$) and $|\mathbf{J}|$ is the determinant of the Jacobian matrix of the trasnformation from $\mathbf{m}_{\rm curr}$ to $\mathbf{m}_{\rm cand}$ (discussed below in Section 2.5). In our algorithm, the generation of candidate models samples the prior distribution as suggested by Mosegaard & Tarantola (1995), so that the proposal and prior distributions are the same and eq. (3) simplifies to

$$\alpha = \min\left[1, \frac{p(\mathbf{d}|\mathbf{m}_{\rm cand},\mathcal{I})}{p(\mathbf{d}|\mathbf{m}_{\rm curr},\mathcal{I})} |\mathbf{J}|\right]. \quad (4)$$

The key function in the acceptance probability of eq. (4) is the likelihood, which quantifies how closely predicted data fits the observed arrival times. The differences between observed and predicted arrival times are typically assumed to be normally distributed and are quantified by a covariance matrix of the errors in the data $\mathbf{C}_{\rm err}$. These errors include noise in the seismic waveforms, differences in the picking of the arrival times by different operators and inaccuracies due to approximations in the forward prediction. In our inverse problem, we assume that the arrival time errors are uncorrelated and all have the same standard deviation $\sigma_{\rm err}$ so that the covariance matrix is diagonal as in $\mathbf{C}_{\rm err} = \sigma_{\rm err}\mathbf{I}$, where $\mathbf{I}$ is the identity matrix. This is a first-order assumption about the data errors widely used in earthquake location (e.g. Sambridge & Kennett 2001) and LET (e.g. Kissling *et al.* 1994). Although errors in the measured traveltimes can be correlated, an in-depth study of these correlations would require an extensive analysis that is beyond the scope of our study. The likelihood function for uncorrelated data errors is

$$p(\mathbf{d}|\mathbf{m},\mathcal{I}) = \frac{1}{\left[(2\pi)^{N_d}\sigma_{\rm err}^{N_d}\right]^{1/2}} \exp\left[-\frac{1}{2}\sum_{i=1}^{N_d}\left(\frac{d_i - g_i(\mathbf{m})}{\sigma_{\rm err}}\right)^2\right]. \quad (5)$$

Following a hierarchical Bayes approach (Malinverno & Briggs 2004), we include the standard deviation of the data errors $\sigma_{\rm err}$ as

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an additional unknown parameter whose PPD can be determined from the data starting from a uniform prior distribution (see the next section). In the calculation of the likelihood, we also weigh the data for station-event distance using a widely adopted scheme (e.g. Paul *et al.* 2001). Unit weight is given to arrival times recorded at stations closer than 40 km to the event, and the weight decreases linearly to zero for station-event distances between 40 and 80 km. The goal is to decrease the effect of large residuals in long ray paths, due to the increase of modelling error with increasing event-station distance.

2.4 Prior probability distribution

In a Bayesian algorithm, prior knowledge about the investigated parameters is described by a prior probability density that represents our initial state of information. Different kinds of prior probability distributions can be used, and both 'hard' and 'soft' priors have been considered (Backus 1988). Here, we quantify prior knowledge in the parameters in **m** with uniform and normal distributions. We assume that the parameters are not correlated *a priori*, so that the full prior distribution $p(\mathbf{m}|\mathcal{I})$ is the product of the prior distributions of each of the components of **m**. We use a uniform prior probability distribution for the number k of elastic nuclei in the subsurface, for the positions of the elastic nuclei **s** in the 3-D volume, and for the standard deviation of the data errors σ_{err}. A normal distribution describes the prior knowledge in the coordinates of the events **e** and in the *P*-wave velocity **v** inside each Voronoi cell. Numerical values of the parameters of the prior distributions (minimum and maximum values for the uniform distributions, mean and standard deviation for the normal distributions) are listed in Table 1.

2.5 RJMCMC implementation

A crucial component of a MCMC algorithm is the 'recipe', the list of moves in parameter space that produces a candidate model for the next step of the chain. As illustrated by Mosegaard & Tarantola (1995), different recipes can be used to solve the same inverse problem. However, a poor choice of the recipe can negatively affect the exploration of the PPD by the sampling chain. Briefly, if the candidate is created with a 'weak' perturbation of the current model, the predicted data will have small changes, and candidates will be often accepted. However, if the candidate changes are very small the chain may converge slowly to sample the PPD. On the other hand, a 'strong' perturbation of the current model will generally result in a candidate whose data fit is significantly worse than that of the current model. These candidates will be typically rejected, slowing down the exploration of the parameter space (Mosegaard 2006).

Our recipe is composed of six different moves. In each of these moves, the perturbations are designed to result in a sequence of candidate models that is distributed as in the prior distribution of the parameters. At each iteration, the candidate model is generated by randomly picking one of the six moves and perturbing the current model according to the following rules:

(1) Perturbation of the standard deviation of the data errors σ_{err} (this move has a probability of 0.10 to be selected).

(2) Perturbation of the position of one event e_i chosen at random. Either the depth $e_{i,z}$ or the horizontal position (longitude $e_{i,x}$ and latitude $e_{i,y}$) is changed (probability 0.15).

(3) Perturbation of the position (x_i, y_i, z_i) of one elastic nucleus chosen at random. These perturbations maintain a minimum distance between two elastic nuclei, computed on the basis of the local density of the fixed 3-D grid of nodes used in forward modelling. Minimum distance must be carefully chosen not to violate detailed balance in the MCMC, see Mosegaard & Sambridge (2002) (probability 0.15).

(4) Perturbation of the *P*-wave velocity V_i of one nucleus chosen at random (probability 0.40).

(5) Birth of an elastic nucleus: one elastic nucleus is randomly added to the current model. The position of the nucleus is randomly picked in the entire 3-D volume. As for move (3), a minimum distance between two nuclei is maintained according to the local density of the fixed grid of nodes. The *P*-wave velocity is randomly picked from the prior probability distribution at the position of the new nucleus (probability 0.10).

(6) Death of a nucleus: one elastic nucleus chosen at random is deleted from the ensemble (probability 0.10).

The balance between 'weak' and 'strong' moves in the above recipe has been set to result in an overall probability of acceptance around 0.25. Numerical experiments done on multivariate normal distributions show that an acceptance probability around 0.25 maximizes

Table 1. Prior distributions of model parameters in vector **m**.

Model parameter	Prior distribution	Prior parameter	Inversion of synthetic data	Inversion of field data
Standard deviation of data errors σ_{err}	Uniform	Minimum	0.03 s	
		Maximum	0.3 s	
Number of elastic nuclei k	Uniform	Minimum	1	
		Maximum	500	
Spatial coordinates of elastic nuclei **s**	Uniform	Minimum	Minimum in model domain	
		Maximum	Maximum in model domain	
P-wave velocity **v**	Normal	Mean	Linear trend between surface (5 km s^{-1}) and 20 km depth (7 km s^{-1})	1-D model of Valoroso *et al.* (2009)
		Standard deviation	Linear trend between surface (1 km s^{-1}) and 20 km depth (0.2 km s^{-1})	Linear trend between surface (1 km s^{-1}) and 20 km depth (0.2 km s^{-1})
Spatial coordinates of events **e**	Normal	Mean	'True' value ±0.5 km	Coordinates in Valoroso *et al.* (2009)
		Standard deviation	1.5 km (Valoroso *et al.* 2009)	

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the efficiency in sampling the parameter space (Gelman *et al.* 1996). We note that the perturbation of the position of a Voronoi nucleus (move 3) is performed in a normalized unit cube. As the extent of our investigated volumes is less in the vertical direction than in the horizontal directions, the absolute nuclei positions are changed by smaller amounts in the vertical than in the horizontal directions. Alternatively, this move could be split in two submoves: one for the perturbation along the horizontal plane, and one for the perturbation along the *Z*-axis.

The acceptance probability of a candidate (eq. 4) contains the determinant of the Jacobian matrix of the transformation from the current to the candidate model. This Jacobian term is relevant if the dimensions of $\mathbf{m}_{\text{curr}}$ and $\mathbf{m}_{\text{cand}}$ are different (the Jacobian equals unity otherwise). For the birth and death moves in (5) and (6) above, the Jacobian can be shown to equal one by following the same reasoning as in appendix B of Piana Agostinetti & Malinverno (2010).

To decide how long to sample, we monitor mean values of the σ_{err} parameter until it becomes approximately constant. We discard the samples previously collected (in the so called 'burn-in' phase) and we run the chain for another $\sim 10^6$ iterations. Repeated trials show that this number of iterations is enough for the chain to sufficiently sample the PPD of the model parameters. Moreover, we run our algorithm on a cluster of 256 CPUs, so that the final sample is composed of the results from several independent chains (30 chains for the synthetic test and 20 chains for the inversion of field data). Each chain is started from an independently obtained sample of the prior distribution, minimizing the influence of the starting point on the final results.

3 INVERSION OF SYNTHETIC DATA

In this section, we present the results of a test that used synthetic *P*-wave arrival times computed in a simple velocity model. The results of this test will demonstrate the main characteristic of the algorithm and its performance in reconstructing subsurface properties.

In the synthetic test, the investigated volume is a parallelepiped with a base of 100 × 100 km and a thickness of 20 km (Fig. 2). The 16 seismic stations are distributed at the surface on a square

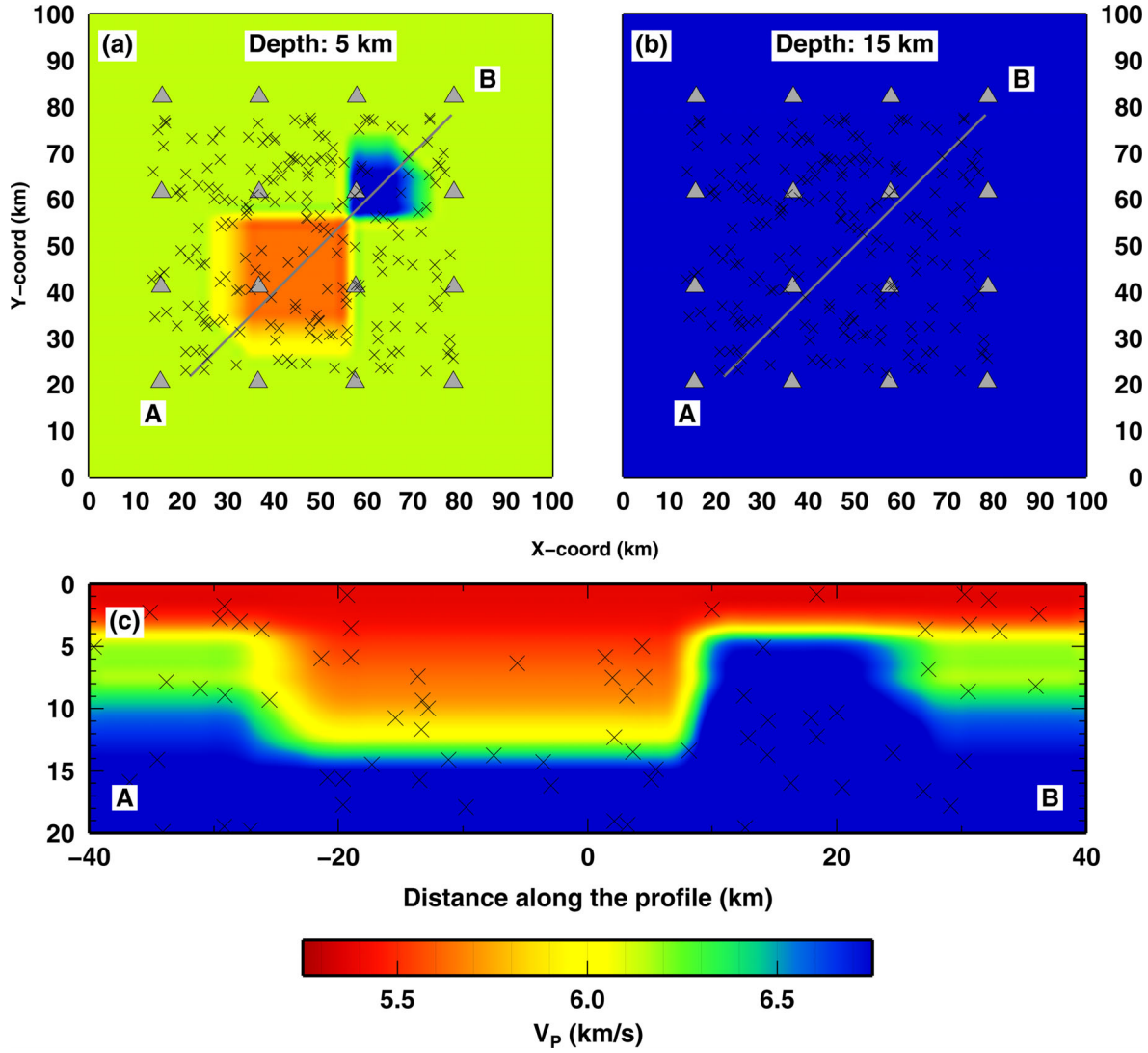


Figure 2. Velocity model used to computed the *P*-wave arrival times in the inversion of synthetic data. (a) Map of the *P*-wave velocity at 5 km depth. Crosses show the event positions and grey triangles the seismic stations in the horizontal plane. The line AB is the trace of the profile in (c). (b) Same as in (a), but at 15 km depth. At this depth, both anomalies in the *P*-velocity model disappear. (c) Cross-section of *P*-wave velocity along the profile AB. Crosses show events occurring within a horizontal distance of 10 km from the profile.

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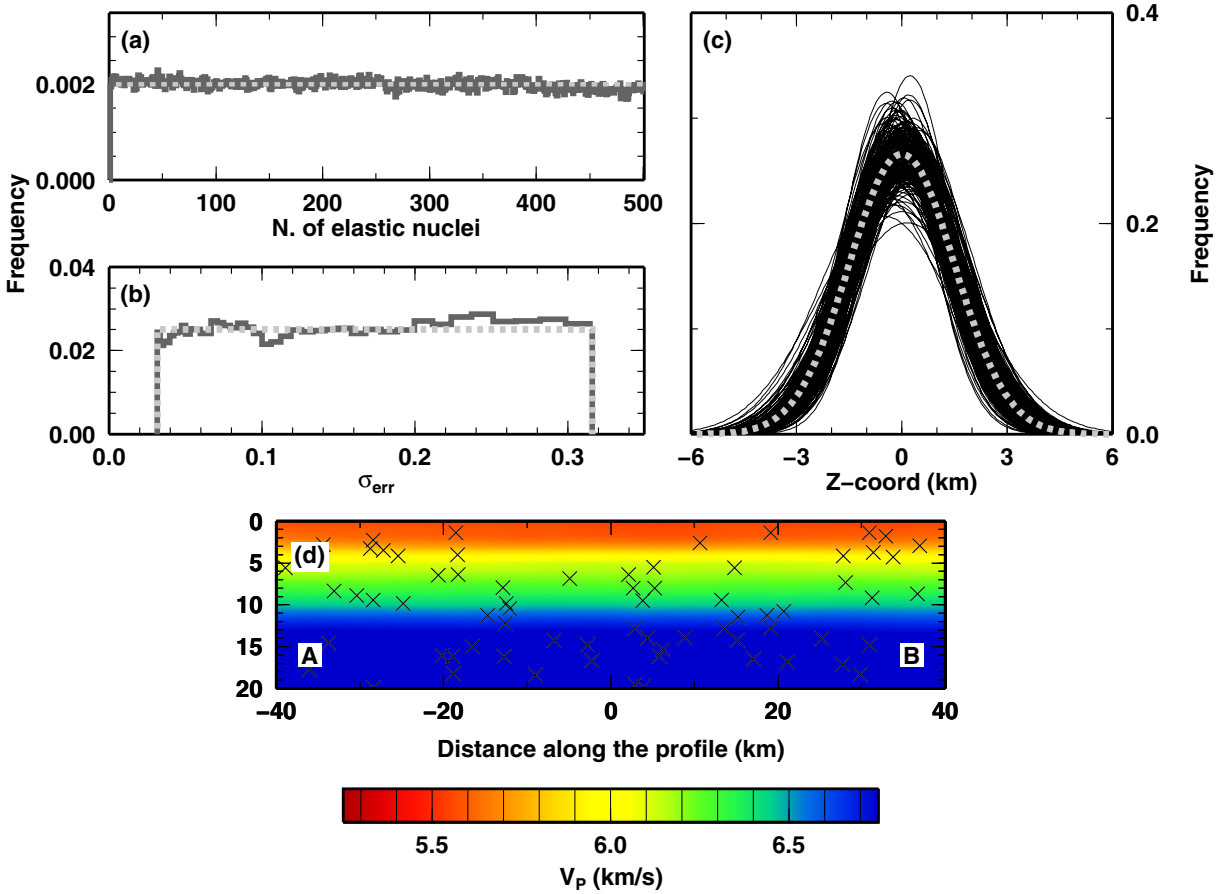


Figure 3. Results of a RJMCMC run where data were ignored and the algorithm sampled the prior distribution. (a) Histogram of sampled values of k (number of elastic nuclei in the investigated volume). The dashed line indicates the uniform prior distribution. (b) As in (a) for the value of the standard deviation of the data errors $\sigma_{\rm err}$. (c) Prior probability distribution of the z-position of the events. For each event, we plot the distribution of the depth difference between the values sampled in the chain and the prior mean value. The dashed line represents the normal prior distribution.(d) Average sampled value of the P-wave velocity at depth, plotted along the profile AB in Fig. 2. Crosses show events occurring within a horizontal distance of 10 km from the profile.

grid, with interstation distance of about 20 km. The position of the 200 events used is randomly generated in the 50 × 50 × 20 km innermost part of the investigated volume. Synthetic P-wave arrival times are computed from each event to each station, generating a data vector of 3200 elements. Gaussian random noise with zero mean and a standard deviation of 0.1 s is added to the synthetic traveltimes. The synthetic model used in the computation (Fig. 2) consists of a background linear velocity increase with depth plus a low-velocity volume 20 × 20 × 10 km from 2.5 to 12.5 km depth and a high-velocity volume 10 × 10 × 10 km in the same depth interval. The value of P-wave velocity in the low (high) anomaly is −10 per cent (+10 per cent) with respect to the background velocity. The 3-D grid for ray tracing has a horizontal spacing of 2.5 km in the centre of the domain, decreasing to 5 km at the edges. The grid spacing is 2.5 km along the vertical axis.

To check that the RJMCMC algorithm samples candidate models distributed as in the prior, we performed a run where all candidates were accepted regardless of the fit to the data. As shown in Fig. 3, the algorithm correctly samples the prior probability distributions of the parameters (Table 1): the number of elastic nuclei in the model and the standard deviation of the data errors are uniformly distributed between their minimum and maximum prior values (Figs 3a and b), the sampled event positions match the normal prior distribution (Fig. 3c), and the prior mean values of P-wave velocity are reproduced (Fig. 3d).

We then show the result of sampling the PPD while taking into account the data with 30 independent chains, which required a total computation time of ∼15 days. Fig. 4(a) shows the evolution during RJMCMC sampling of the average misfit to the data and the number of elastic nuclei in each model. The grey area indicates the burn-in phase (i.e. the first 9 × 10^5 models of each chain are not used to compute the PPDs). In the burn-in phase, the algorithm increases the number of elastic nuclei to improve the overall fit to the data. After the burn-in phase, a stable average misfit is reached and the algorithm slowly decreases the number of elastic nuclei in each chain, removing model complexity that is not supported by the data. Although the number of elastic nuclei in a chain occasionally increases during the sampling, the mean value does not, demonstrating the parsimony of the algorithm. Fig. 4(b) shows the histogram of the sampled number of elastic nuclei. The number of elastic nuclei is between 50 and 170, with a mode at about 100. Fig. 4(c) displays the histograms of the number of elastic nuclei immediately after the end of the burn-in phase and at the end of the MCMC sampling. The close resemblance between these two histograms confirms the convergence of the algorithm after the burn-in phase.

Fig. 5 illustrates the PPD of P-wave velocity in the investigated volume. As in Fig. 2, we display the velocity maps at two different depth levels (5 and 15 km) and a velocity cross-section through the model. The velocities displayed are the posterior mean values.

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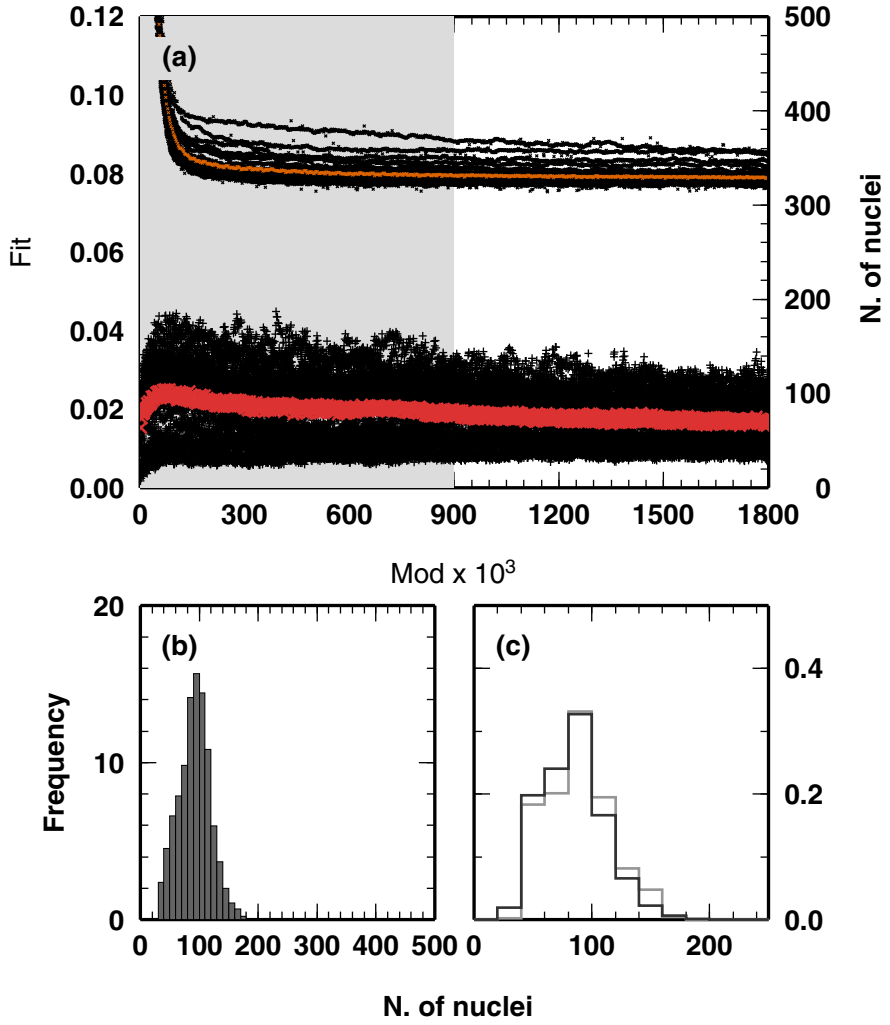


Figure 4. Evolution of parameters during RJMCMC sampling using synthetic data, for the 30 independent chains. (a) Black dots are the mean rms misfit to the data, and crosses display the number of elastic nuclei in each chain. Red and orange crosses show the average of the misfit and the number of elastic nuclei, respectively. Each symbol is the 10 000th sampled model along each chain. The grey area indicates the burn-in phase. (b) Histogram of the number of elastic nuclei in the models sampled after the burn-in phase. (c) Histograms of the number of elastic nuclei in the models sampled. Light (dark) grey line represents the first (last) 10^5 models after the burn-in phase.

Plotting the posterior mean is one way to display the sampling results; as our algorithm samples the full PPD, however, other estimators could be displayed as well, for example the posterior median (see Bodin *et al.* 2012, for a discussion). An important feature of the PPD that quantifies the reliability of the inversion is the posterior marginal uncertainty, the uncertainty in any single parameter given all the values that the other parameters can take. We visualize these posterior marginal uncertainties with the colour intensity, similarly to Oldenburg & Li (1999). The posterior mean velocities are plotted in Fig. 5 with full colour saturation if the posterior standard deviation is $\leq$20 per cent of the prior standard deviation. The colour saturation decreases linearly to zero where the posterior standard deviation is $\geq$50 per cent of the prior standard deviation. This simple display highlights the regions where the data closely constrain the solution.

Fig. 5(a) shows that the top of the two velocity anomalies is accurately reconstructed. In this synthetic test, the overall geometry of the two velocity anomalies is closely reconstructed because of the homogeneous distribution of the event-station pairs. On the other hand, the absolute value of the high-velocity anomaly at 5 km depth is less well constrained, probably due to a relatively low number of rays crossing the top of this anomaly. Fig. 5(b) shows that the bottom of the low-velocity anomaly is also well defined. In this case, the arrival times of the deeper events (focal depth larger than 15 km) contain enough information to delimit the depth extension of the low-velocity anomaly. Finally, the vertical boundaries of the velocity anomalies are correctly reconstructed, as seen in Fig. 5(c). The velocity model at 15 km depth is less resolved at the northeastern end of the profile, where no event deeper than 10 km is present, compared to the southwest, where there are deep events.

Fig. 6(a) illustrates the differences between the posterior mean model and the 'true' model. Velocity values in the two anomalies are slightly overestimated, and such differences are compensated by an underestimation of velocity in the surrounding area. These differences are significant only for the negative velocity anomaly. Fig. 6(b), which plots the ratio between the absolute velocity differences between the two models and the posterior standard deviation, shows that only the negative anomaly has a posterior mean *P*-wave velocity that differs from the 'true' model by more than one standard deviation.

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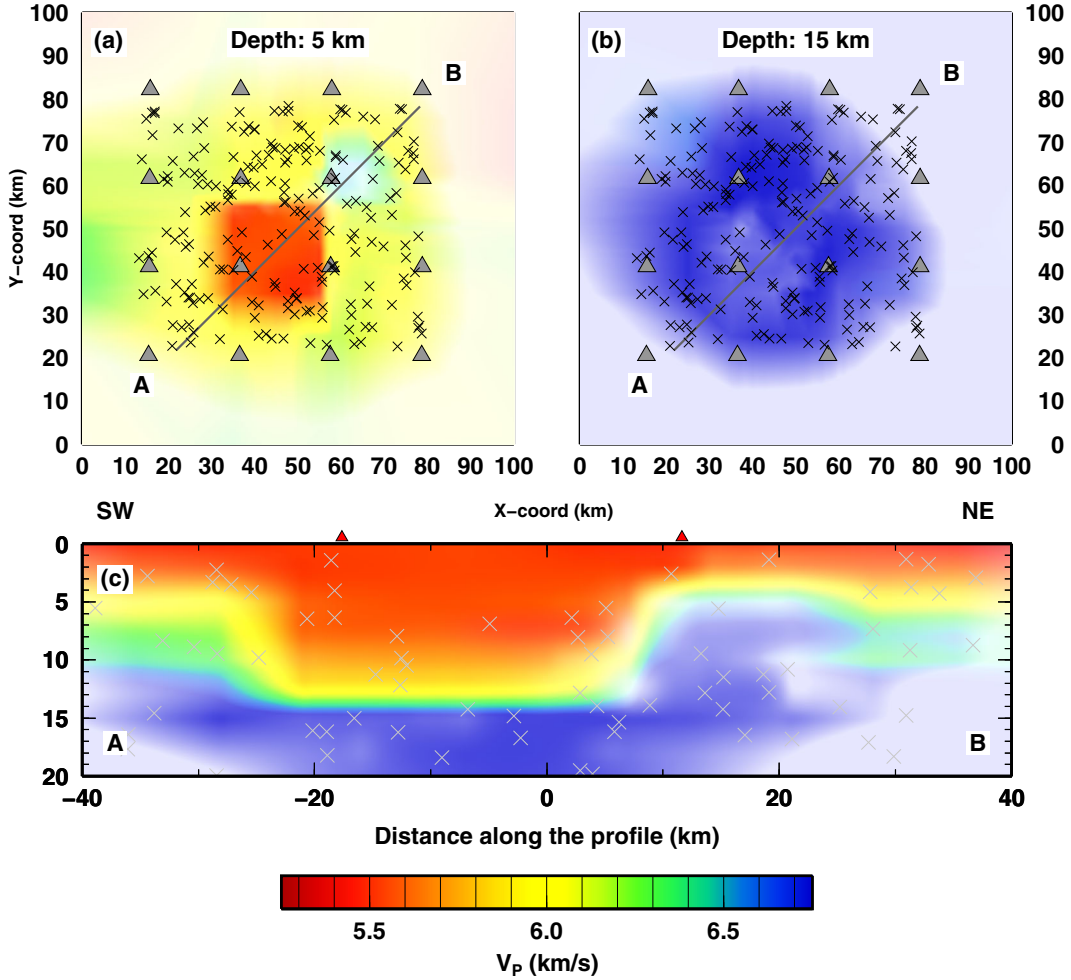


Figure 5. *P*-wave velocity model reconstructed after RJMCMC sampling in the inversion of synthetic data. In each panel, the intensity of the colours depends on the ratio of the posterior to the prior standard deviation (see the text). (a) Map of the posterior mean of *P*-wave velocity at 5 km depth. Crosses show the event positions and grey triangles the seismic stations in the horizontal plane. The line AB is the trace of the profile in (c). (b) Same as in (a), but at 15 km depth. (c) Cross-section of posterior mean of *P*-wave velocity along the profile AB. Crosses show events occurring within a horizontal distance of 10 km from the profile. Red triangles display the position of seismic stations along the profile.

Fig. 7 illustrates how the algorithm performs in reconstructing the coordinates (location and origin time) of the seismic events. In each panel, we plot the difference between the posterior mean value and the 'true' value of its *X*-, *Y*-, *Z*- and *t*-position. Thus, for a perfect sampling, all the symbols should plot along the dashed white line, which marks a zero difference between posterior mean and 'true' value. To demonstrate that the algorithm can deal with an initial bias for event locations and origin times, the prior means were displaced by 0.5 km (*X*, *Y* and *Z*) and 0.5 s (*t*) from their actual values. These biased prior means are shown by dotted white lines in Fig. 7. Both *X*- and *Y*-positions are fairly well recovered by the algorithm (Figs 7a and b), while the depth *Z* is not as well constrained. However, the plot for the *Z*-position correctly indicates a shift towards the true value from the biased prior mean for almost all the events (Fig. 7c). Finally, Fig. 7(d) shows that the *t*-positions of the events, reconstructed with the 'minimum origin time' criterion (Section 2.2), are closely constrained by the data.

The fit between observed and synthetic arrival times (residuals) is plotted in Fig. 8. The residuals are in agreement with the white noise added to the arrival times computed through the 'true' model. This conclusion is confirmed by the histogram of the residuals (Fig. 8b), which approximates a normal distribution with zero mean and a standard deviation of about 0.1 s.

4 INVERSION OF FIELD DATA

In this section, we present the results from the inversion of field measurements recorded during a temporary deployment in the Val D'Agri region, Southern Italy (Valoroso *et al.* 2009). The data set consists of about 9700 *P*-wave arrival times from 772 events, measured by about 35 seismic stations that operated in the area between 2009 May and 2009 August. Event selection and *P*-wave arrival time picking are described in Valoroso *et al.* (2009). In this study, we use as prior mean values the event locations and the best 1-D velocity model of Valoroso *et al.* (2011). In particular, the prior standard deviations in the spatial coordinates of the events (1.5 km for both the horizontals and the vertical) are consistent with the dense station spacing of this data set (Valoroso *et al.* 2009). The effect of these parameters on the PPD has been investigated and the results are presented in the Appendix. Other parameters of the prior are listed in Table 1.

The investigated volume covers a 100 × 100 km square extending to 20 km depth. For ray tracing, we use a regular 3-D grid of

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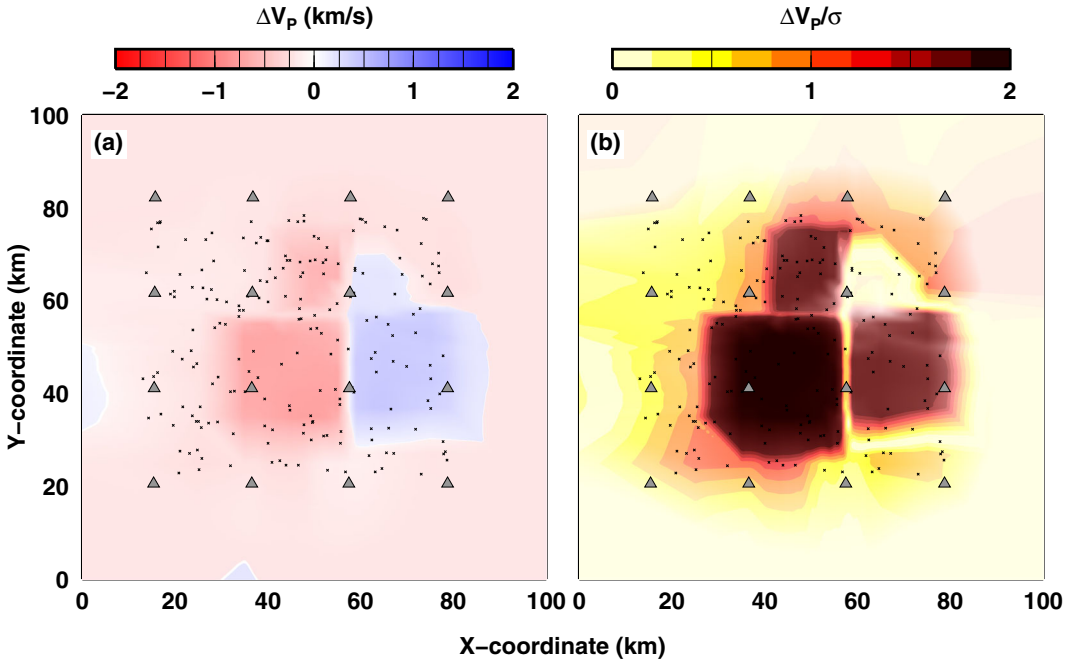


Figure 6. (a) Differences between the posterior mean and the 'true' models at 5 km depth. The intensity of the colours depends on the ratio of the posterior to the prior standard deviation as in Fig. 5 (see the text). (b) Ratio between the absolute value of the velocity differences in (a) and the posterior standard deviation of velocity at 5 km depth.

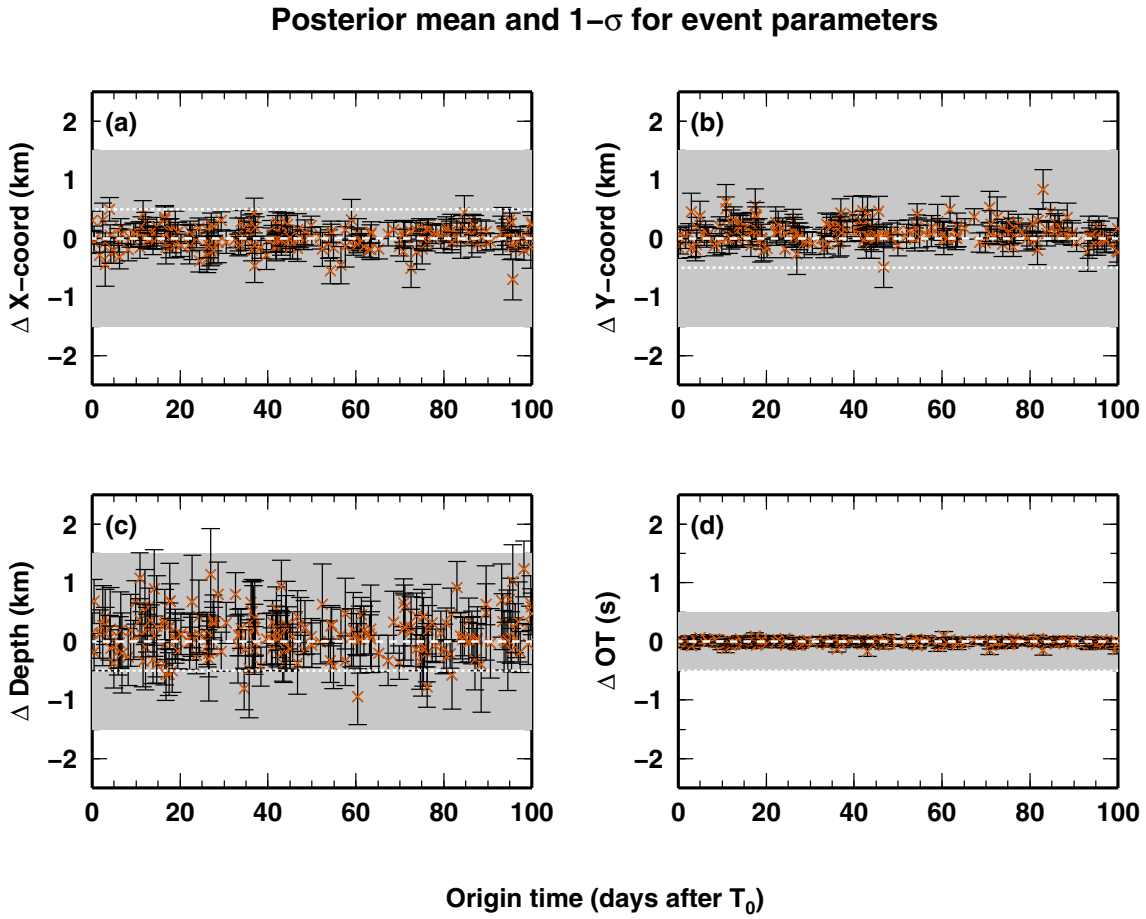


Figure 7. Posterior mean value and standard deviation of the event coordinates in the inversion of synthetic data. Each panel plots the difference between the 'true' value and the posterior mean. White dotted lines indicate the difference between the prior mean and the 'true' values. White dashed lines mark a zero difference. The grey area indicates the prior standard deviation. The panels show the differences for (a) longitude, (b) latitude, (c) depth, and (d) origin time.

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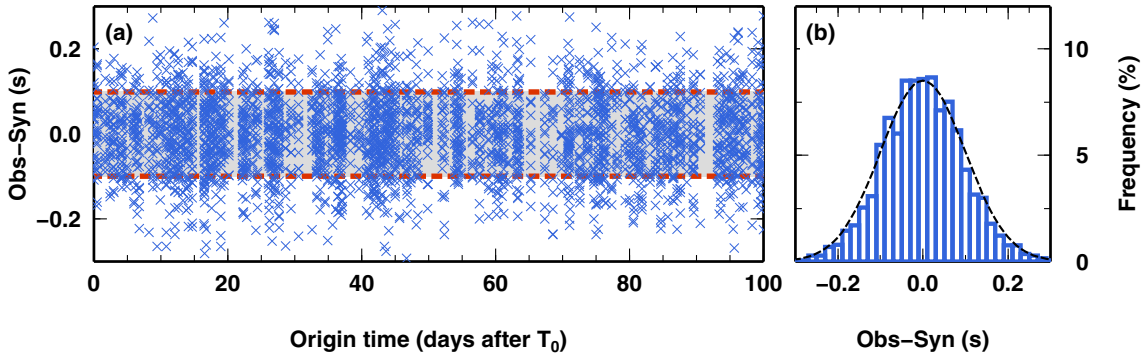


Figure 8. Residuals defined as differences between observed and posterior mean values of the *P*-wave arrival times in the inversion of synthetic data. (a) Residuals are shown by blue crosses. Red dashed lines indicate the standard deviation of the residuals, which should be similar to the standard deviation of the white noise added to the synthetic data (0.1 s). (b) Histogram of the residuals in (a). A grey dashed line indicates a theoretical gaussian distribution for the residuals.

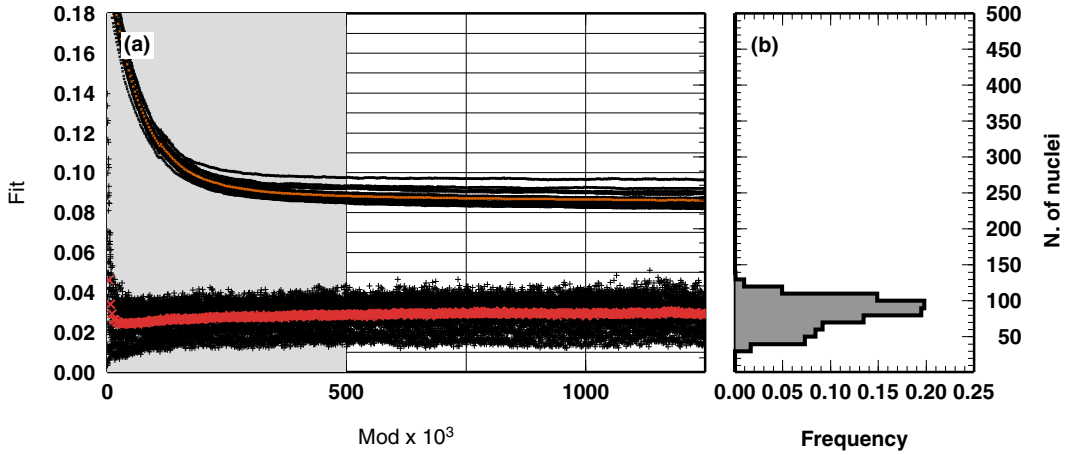


Figure 9. Evolution of parameters of 20 chains during RJMCMC sampling using field measurements. (a) Black dots are the mean rms misfit to the data, and circles display the number of elastic nuclei in each chain. Red and orange crosses show the average in the five chains of the misfit and the number of elastic nuclei, respectively. Each symbol is the 10 000th sampled model along each chain. The grey area indicates the burn-in phase. (b) Histogram of the number of elastic nuclei in the models sampled after the burn-in phase.

$19 \times 19 \times 12$ nodes spaced every 3 km in the central part and every 10 km along the edge of the domain. The RJMCMC sampling consists of 20 independent chains that sampled 1.25×10^6 models each. The 'burn-in' period is 5×10^5, so that the final PPDs are reconstructed from about 15×10^6 velocity models. Total run-time was about 30 d. The results obtained with the RJMCMC algorithm will be compared to the inversion results obtained by Valoroso *et al.* (2011) using the *SimulPs* code (Evans *et al.* 1994).

In Fig. 9(a), we plot the evolution of the data misfit together with the number of elastic nuclei. The misfit decreases rapidly in the first few 10^5 models and becomes nearly constant after the end of the burn-in phase. The number of elastic nuclei in the model is very variable in the first 10^4 models, where the sampled models have as few as 10 or as many as 400 elastic nuclei. The number of elastic nuclei then becomes stable around 100. The histogram of the number of elastic nuclei for models sampled after the burn-in phase (Fig. 9b) displays a single mode between 80 and 100 nuclei.

Fig. 10 compares *P*-wave velocity models computed using the two inversion techniques at 3 km depth. Fig. 10(a) displays the posterior mean *P*-wave velocity obtained using our RJMCMC algorithm, while Fig. 10(b) plots the results of the *SimulPs* inversion (see Valoroso *et al.* 2011, for details). As done in the synthetic data inversion, the colour intensity in Fig. 10(a) indicates the posterior uncertainty. The posterior mean velocities are plotted with full colour saturation if the posterior standard deviation is $\leq$20 per cent of the prior standard deviation. The colour saturation decreases linearly reaching zero where the posterior standard deviation is $\geq$50 per cent of the prior standard deviation. This display is useful to highlight the well-resolved regions and to de-emphasize features in poorly resolved areas.

The label 'ACP (Apulian carbonate platform)' in Fig. 10 indicates two high-velocity areas where the ACP has been mapped to 3 km depth using active seismic data (Valoroso *et al.* 2011). The two inversion results agree with the general NW–SE trend of these high-velocity structures and with the location of the ACP at this depth level (corresponding to regions with a *P*-wave velocity of 6 km s^{-1} or higher). However, the absolute values of the high-velocity anomalies are different in the two inverted models. In the RJMCMC posterior mean model, the average *P*-wave velocity at 3 km depth is about 0.5 km s^{-1} lower than in the *SimulPs* model. This difference is likely due to a relatively small number of seismic events in the 0–4 km depth range.

Posterior means and standard deviations for the space and time event coordinates are shown in Fig. 11 as differences between the posterior and prior means. These coordinates are generally well resolved from RJMCMC sampling for almost all the events (i.e. the posterior standard deviation is less than the prior). However, we note that for some events the difference between posterior

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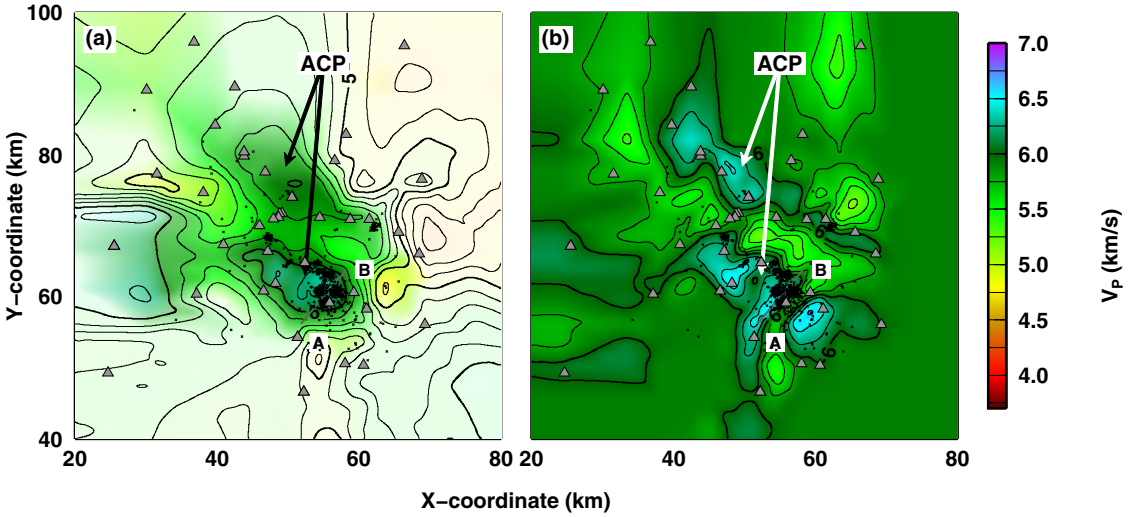


Figure 10. Comparison of *P*-wave velocity models obtained with two tomographic inversion techniques from the same field measurements. Both maps plot *P*-wave velocity at 3 km depth. Crosses show the event positions and grey triangles the seismic stations in the horizontal plane. The line AB is the trace of the profile in Fig. 12. Arrows indicate the Apulian carbonate platform (ACP). (a) Posterior mean of *P*-wave velocity from RJMCMC sampling. The intensity of the colours depends on the ratio of the posterior to the prior standard deviation as in Fig. 5 (see the text). (b) *P*-wave velocity obtained using the *SimulPs* code.

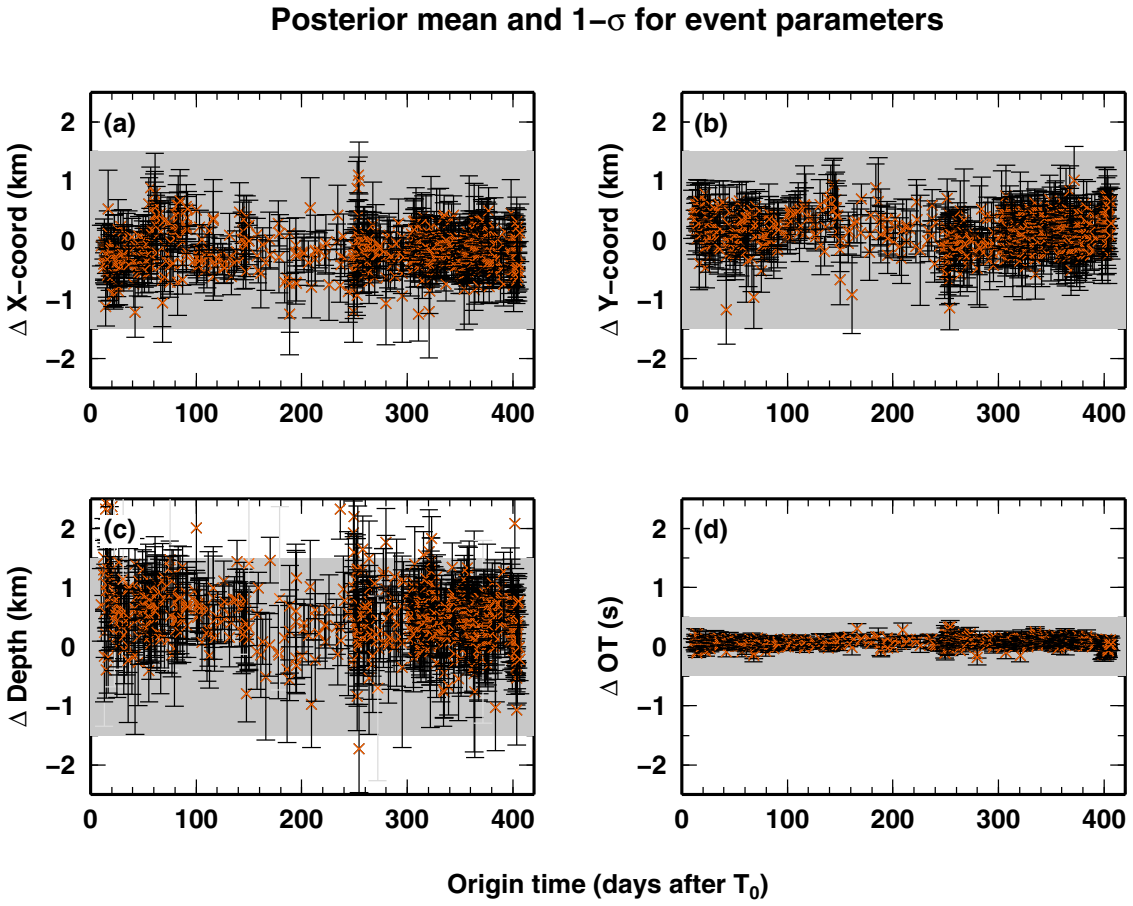


Figure 11. Posterior mean value and standard deviation of the event coordinates in the inversion of field measurements. Each panel plots the difference between the prior and the posterior mean. The grey area indicates the prior standard deviation. The panels show the differences for (a) longitude, (b) latitude, (c) depth and (d) origin time. Grey error bars indicate event coordinates where the posterior standard deviation is similar to the prior.

and prior mean can be large (more than 1 km), especially for the depth of the events (as great as 2 km). As noted above, the two inverted models seem to have different average velocities, with the RJMCMC model slightly slower than *SimulPs* model. This difference can shift the hypocentres, on average, towards greater depths.

To further evaluate the performance of the RJMCMC algorithm, we plot the posterior position of the hypocentres along a profile

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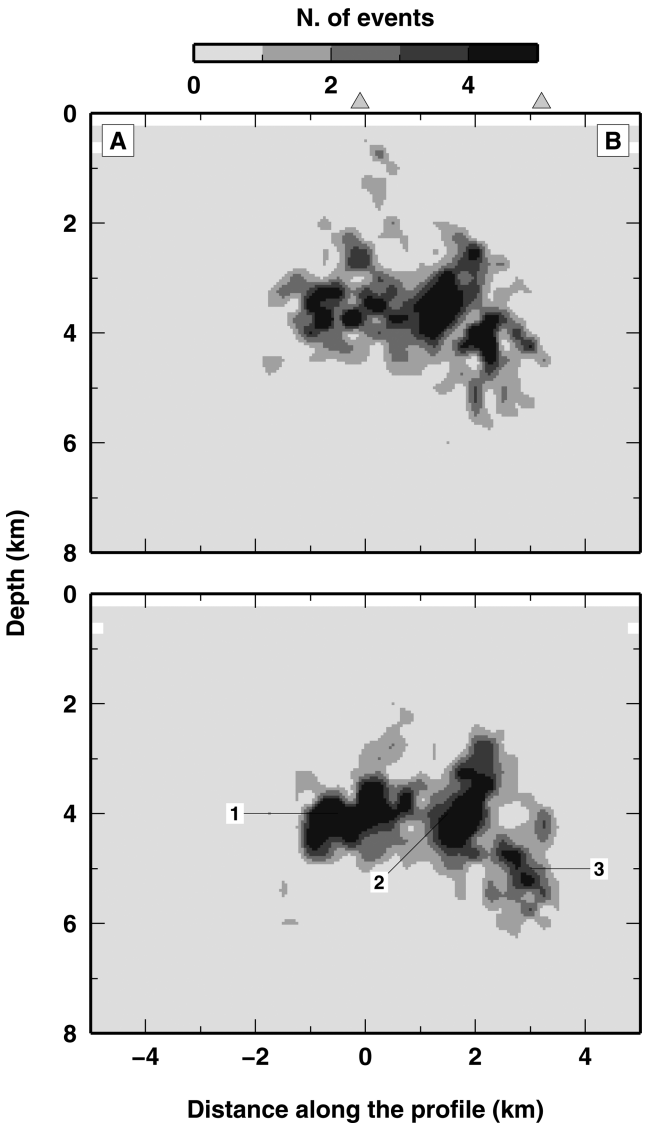


Figure 12. Prior and posterior mean position for the events used in the inversion, projected along line AB in Fig. 10. Only events 10 km or closer to the cross-section are considered. (Upper panel) Density of event locations in the *SimulPs* inversion of Valoroso *et al.* (2011). (Lower panel) Densities of the posterior mean event locations in our RJMCMC sampling. Numbers 1–3 refer to cluster of events.

across the region with most recorded seismic activity (Fig. 12). In the upper panel, we show the position of the hypocentres obtained by Valoroso *et al.* (2011), which form a number of clusters at about 4 km depth. The lower panel displays the posterior positions of the hypocentres from the RJMCMC sampling. The overall hypocentral positions are similar in the two inversions. However, RJMCMC sampling results in a separation of the events in three elongated clusters that dip towards the southwest.

Fig. 13 shows the differences between the observed *P*-wave arrival times and their posterior mean ('residuals' hereafter). The posterior mean of the arrival times is computed as the average of the arrival times obtained for all the velocity models sampled by RJMCMC. As the relationship between arrival times and velocity is nonlinear, these posterior mean arrival times are not necessarily the same as the arrival times computed for the posterior mean of the velocity model (the model in Fig. 10). The standard deviation of the residuals is about 0.12 s, which is in agreement with the rms residual of 0.14 s obtained in the *SimulPs* inversion of Valoroso *et al.* (2011).

5 DISCUSSION

Based on the results presented in the previous section, we discuss some key characteristics of our Monte Carlo approach to the LET inverse problem, that is its intrinsic parsimony, the ability to determine if and where the data improve our prior knowledge, and the possibility to determine the posterior correlation between the investigated parameters. Finally, we include some considerations on the *P*-wave velocity values and the event locations obtained by our algorithm.

5.1 Parsimony of the solution

An attractive feature of Bayesian inference is its intrinsic parsimony: given two hypotheses that fit the data equally well, the hypothesis with fewer adjustable parameters will have a higher posterior probability (Gull 1988; Jefferys & Berger 1992; Sivia 1996; Jaynes 2003). This feature has been used in geophysical inversion to obtain earth models that have as few parameters as needed to fit the data (e.g. Malinverno 2000, 2002; Malinverno & Leaney 2005; Sambridge *et al.* 2006; Piana Agostinetti & Malinverno 2010; Sambridge *et al.* 2013).

This parsimonious characteristic is also present in our RJMCMC algorithm, as demonstrated by the inversion results for both synthetic data and field measurements. Figs 4 and 9 show that the number of elastic nuclei in the sampled velocity models does not increase during sampling to improve the fit to the data. In both inversions, the most probable number of nuclei is about 100, approximately 1/15 of the number of nodes in the fixed 3-D grid used for ray tracing. Thus, the RJMCMC algorithm focuses sampling on models that have a limited number of parameters and fit the data to some degree. The level of fit to the data, as shown in Figs 8 and 13, is also determined by the algorithm itself through the parameter σ_{err}. The intrinsic parsimony of the velocity model avoids the need for a user-determined smoothing parameter.

5.2 Comparing posterior and prior knowledge

Although a Monte Carlo inversion is computationally expensive, it provides an extremely useful quantification of the posterior uncertainties. An accurate assessment of the posterior uncertainties in tomographic inversion is essential to evaluate the regions of the investigated volume that have been resolved by the data and to compare the recovered seismic model with information obtained from different sources (e.g. other tomographic images or well-logs). In the Bayesian framework we adopt, the well-resolved regions are those where our posterior knowledge is a significant improvement over the prior. In our examples, where prior information on *P*-wave velocity is given in terms of a normal distribution, this improvement can be assessed by comparing the posterior and prior standard deviations of the velocity. Regions that are well resolved should have a posterior standard deviation that is clearly less than the prior.

Our test using synthetic measurements illustrates this concept. Fig. 5 shows the effect of having no information (i.e. no events and no seismic stations) in the outer part of the investigated volume, where the posterior standard deviation of the *P*-wave velocity equals the prior. In the cross-section in Fig. 5(c), the deep corner on the northeast side contains no seismic events and the algorithm correctly

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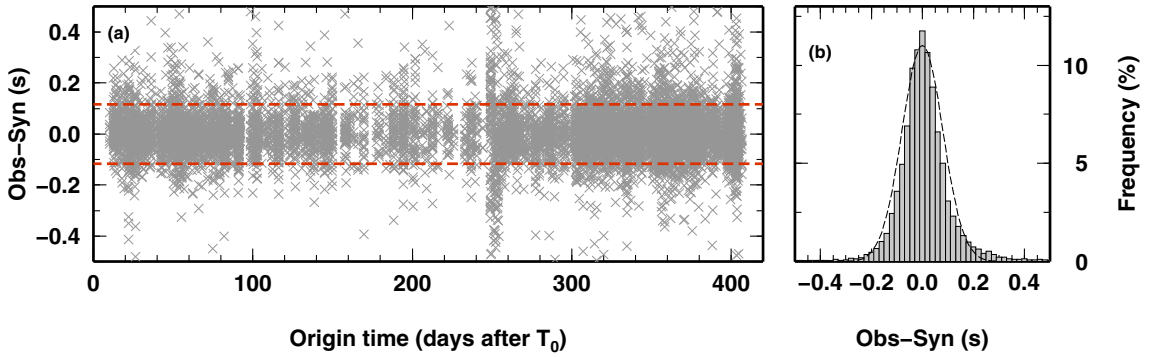


Figure 13. Residuals defined as differences between observed and posterior mean values of the *P*-wave arrival times in the inversion of field measurements. (a) Residuals are shown by grey crosses. Red dashed lines indicate the standard deviation of the residuals (0.12 s). (b) Histogram of the residuals in (a). A grey dashed line indicates a theoretical gaussian distribution for the residuals and shows that our assumptions on error statistics are correct.

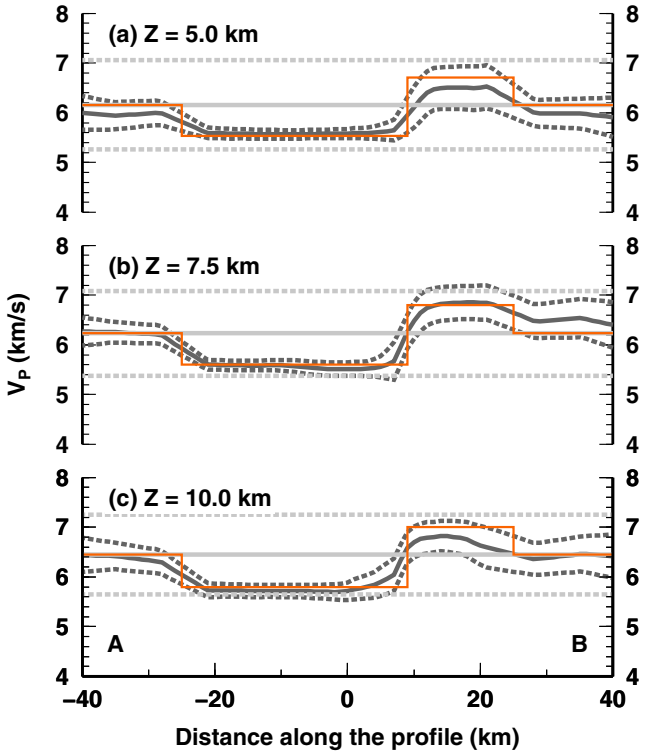


Figure 14. *P*-wave velocity at different depth levels along profile AB, in the synthetic test inversion. Dark (light) grey lines indicate values from the posterior (prior) sampling. Dashed lines show 1σ confidence intervals. Continuous grey lines indicate mean values. An orange line displays the 'true' value. (a) 5.0 km depth. (b) 7.5 km depth. (c) 10.0 km depth.

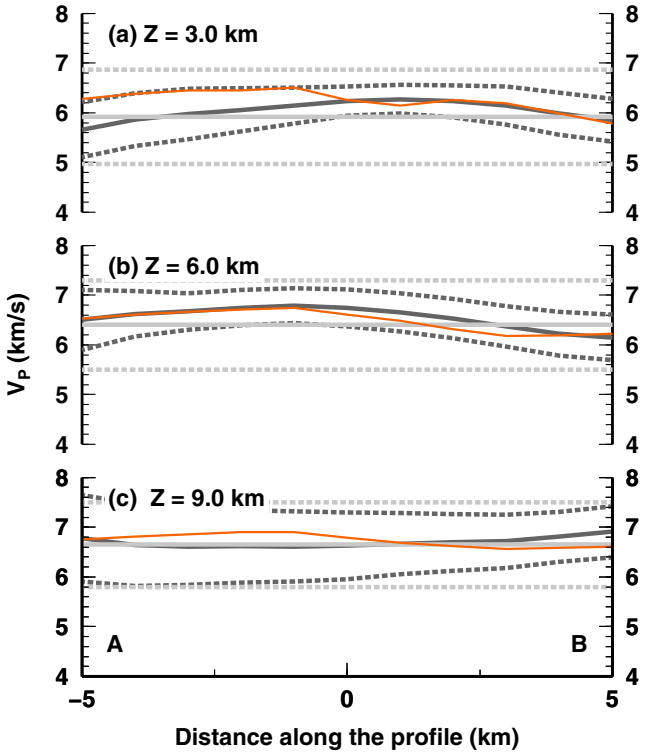


Figure 15. *P*-wave velocity at different depth levels along profile AB, in the inversion of field measurements. Dark (light) grey lines indicate values from the posterior (prior) sampling. Dashed lines show 1σ confidence intervals. Continuous grey lines indicate mean values. An orange line displays the value in *SimulPs* inversion. (a) 3.0 km depth. (b) 6.0 km depth. (c) 9.0 km depth.

results in a posterior standard deviation that equals the prior. In the inversion of field measurements, the well-resolved region is restricted to the central part of the investigated volume, where most stations and events are located (Fig. 10). This region corresponds to the area where we observe the most detailed structure in the *SimulPs* model.

The posterior uncertainties of the reconstructed *P*-wave velocities are further illustrated along cross-sections in Figs 14 and 15. The inversion for the synthetic data (Fig. 14) gives a realistic quantification of the uncertainties, as the posterior mean is almost everywhere within one posterior standard deviation of the true model. The results of our RJMCMC inversion are compared to the *SimulPs* inverted model in Fig. 15. The differences between the posterior mean we obtain and the *SimulPs* inversion are generally within one posterior standard deviation, showing that these differences are within the overall uncertainty of the inversion and hence not significant. An interesting result is that in the deepest part of the cross-section ($Z = 9$ km, Fig. 15c) the posterior uncertainties approach the prior, showing that the data do not constrain this portion of the investigated volume.

For the event coordinates, we generally observe posterior standard deviations that are smaller than the prior (Figs 7 and 11). In the test using the field measurements, these posterior standard deviations decrease to as little as about half the prior value for the *x*- and *y*-coordinates, while they are larger for the *z*-coordinate. In some

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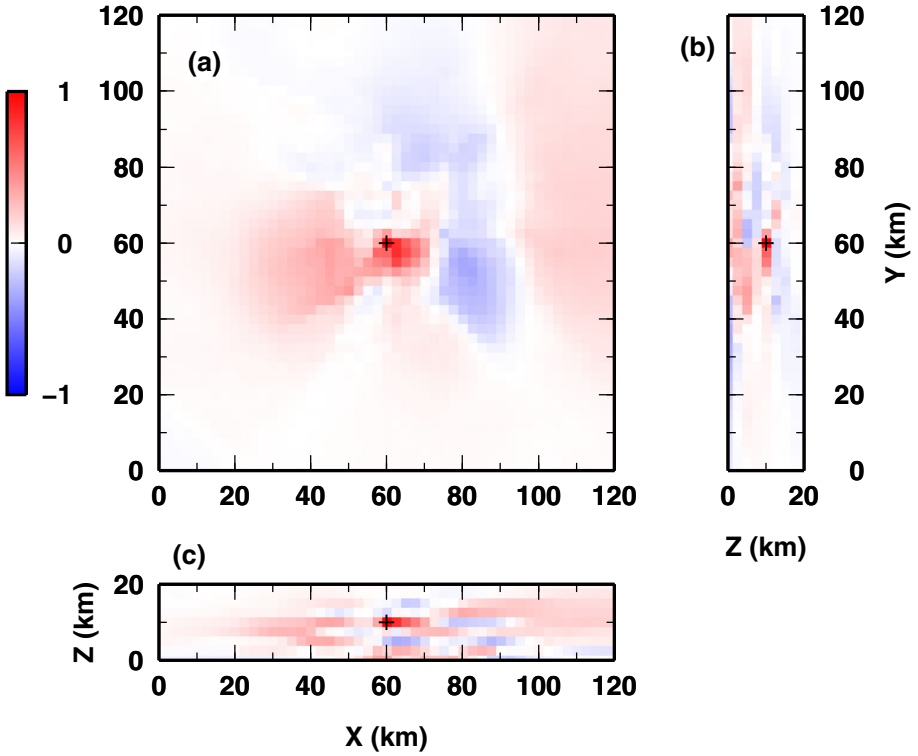


Figure 16. Correlation coefficients between the *P*-wave velocity value in the central node of the investigated volume and the *P*-wave velocity value in each node on three orthogonal planes cutting the volume in its centre. Colours represent the correlation coefficient (i.e. perfect correlation is shown in red, perfect anticorrelation in blue). Black crosses show the position of the central node on the three planes. (a) Correlation coefficients on the *XY* plane. (b) Correlation coefficients along the *YZ* plane. (c) correlation coefficients along the *XZ* plane.

cases, the *z*-coordinate is not resolved (i.e. the posterior standard deviation equals the prior). This reflects the intrinsic difficulty to resolve focal depths using *P*-wave arrivals only and can be improved by using both *P*- and *S*-wave arrival times.

5.3 Correlation between investigated parameters

An advantage of sampling the PPD is that the posterior correlation can be easily computed for any combination of model parameters. This posterior correlation would be hard or impossible to estimate with other inversion methods. To illustrate this feature, we compute for the field data inversion the posterior correlation coefficients between the V_P value in the central node of the 3-D grid and the V_P value in all the nodes in three orthogonal planes passing through the central node (Fig. 16). Clearly, any other combination of parameters could be selected to investigate specific details of the results.

The correlation coefficient is computed from the values of the two parameters in the ensemble of models sampled from the PPD. In our implementation, we save only 1/1000 of the sampled models for a total of about 1.5×10^4 models. In Fig. 16(a), we display the posterior correlation coefficients between the V_P values sampled in the central node of our 3-D grid and the V_P values in all the other nodes on the horizontal plane at 10 km depth. In Figs 16(b) and (c), we show the correlations coefficients along two orthogonal sections, N–S and E–W, passing through the central point of the 3-D grid (i.e. $X = 60$ km and $Y = 60$ km).

The size of the bright red region with high positive correlations around the central point (at about $X = 60$–70 km, $Y = 50$–60 km) shows the overall spatial resolution of the inversion. Fig. 16(a) also shows a broad red region west of the central node (at about $X = 20$–70 km, $Y = 40$–70 km) that is positively correlated to the central node and a blue area to the east (at about $X = 80$–100 km, $Y = 40$–90 km) that is anticorrelated. This result implies that an increase in the velocity of the central node would require an increased velocity in the western red region and a concomitant decrease in the eastern blue region to maintain the fit to the data. Figs 16(b) and (c) display more small-scale posterior correlations induced by the data between the values of V_P in the central node and all the other nodes in the sections.

5.4 Comparing absolute *P*-wave velocities

The absolute value of *P*-wave velocities at depth is always difficult to recover in a linearized tomographic inversion, and can be strongly influenced by the choice of reference model (Kissling *et al.* 1994). In our synthetic data test, the absolute values of the two *P*-wave velocity anomalies are well resolved both at shallow depth and at their bottom (see Figs 2c and 5c). The top of the positive anomaly seems to be not as well resolved as the top of the negative one (Fig. 5), probably because of the small number of events near the top of the positive anomaly. The bottom of both anomalies is quite well resolved because there are a number of events beneath the anomalies.

In the inversion of field measurements, we noted a difference in the absolute *P*-wave velocities found with the two inversion methods over two high-velocity regions (Figs 10a–b). These positive anomalies can be linked to the top of the ACP, a multilayer carbonate platform involved in the Apenninic orogeny. A value of about

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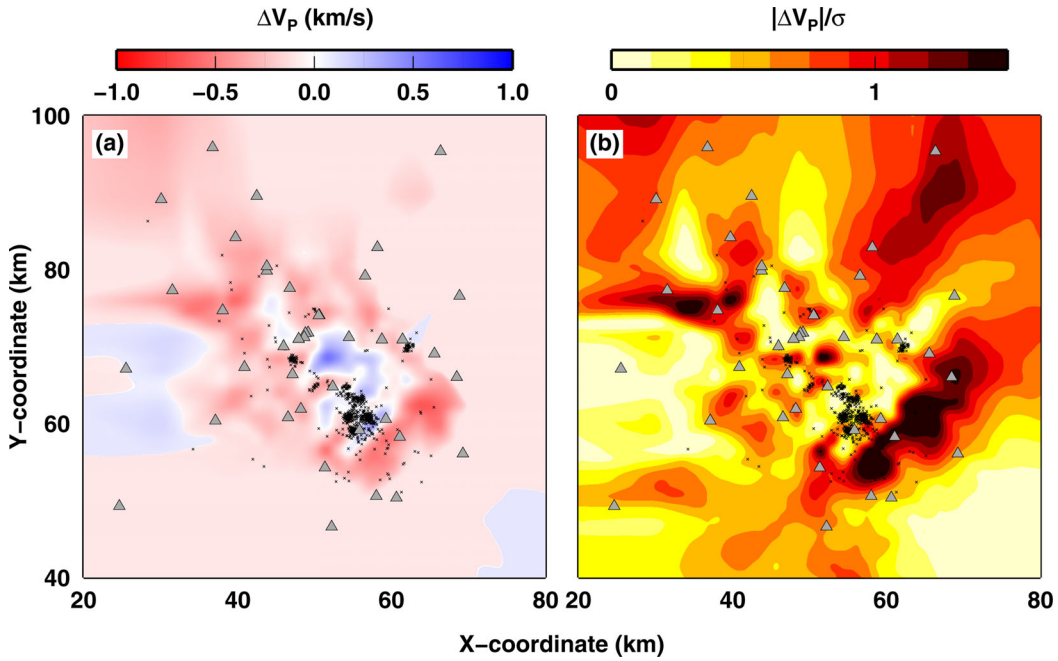


Figure 17. Comparison of differences between the RJMCMC and *SimulPs* inversions at 3 km depth. (a) Difference between the posterior mean of *P*-wave velocity from RJMCMC sampling and the velocity inverted by *SimulPs*. (b) Absolute velocity differences in (a) divided by the posterior standard deviation of *P*-wave velocity from RJMCMC sampling, showing that most differences are within one posterior standard deviation.

6.0 km s^{-1}, found in the RJMCMC inversion, is expected for the top layer of the platform (Improta *et al.* 2000), whereas the lowermost section of the platform (Burano formation) should have a higher velocity. Thus, the results obtained by the RJMCMC algorithm are in agreement with both the location of the top of the ACP from active seismic studies and with the *P*-wave velocities expected in such carbonate rocks.

How significant are the differences in *P*-wave velocity estimated with the two inversion methods? The posterior uncertainties quantified by Monte Carlo sampling allow for an informed comparison. Fig. 17(a) plots the differences between the RJMCMC posterior mean and the *P*-wave velocity obtained by *SimulPs* at 3 km depth. Most of the differences are negative because the RJMCMC inversion gives generally slower values, as noted earlier. The significance of these differences can be assessed by comparing them to the RJMCMC posterior standard deviations. The absolute *P*-wave velocity differences are generally less than one posterior standard deviation (Fig. 17b), meaning that they are broadly within the uncertainty inherent to the inversion. Therefore, the results of the two methods are not significantly different.

5.5 Estimating hypocentre coordinates

Seismic events are generally clustered along tectonically active structures (i.e. fault planes), so that hypocentre coordinates are considered to be well resolved when events are imaged as dipping planes. In general, tomographic inversion is not best suited for hypocentre determination and more specific codes have been developed for such tasks (e.g. Waldhauser 2001). Also, hypocentre estimation usually relies on a 1-D model reconstructed using a limited subset of the recorded data (Kissling *et al.* 1994), which can bias the determination of the focal depth. In our inversion of synthetic data, events were randomly distributed in the central region of the investigated volumes, and thus we cannot evaluate whether the algorithm can cluster events. Nonetheless, we observe in Fig. 7 that the algorithm removes the bias that was added to the event coordinates, even if the focal depths have generally larger posterior standard deviation than the horizontal coordinates. In the inversion of field measurements, we note that the clusters of seismicity obtained with the RJMCMC algorithm define three coherent southwest-dipping structures (Fig. 12). This dip direction is in good agreement with local tectonic structures outlined by geomorphology and active seismics (Valoroso *et al.* 2011), suggesting that RJMCMC sampling resulted in improved hypocentre locations.

6 CONCLUSIONS

We applied a trans-dimensional Monte Carlo algorithm to the LET inverse problem, where event arrival times are inverted to solve for the spatial distribution of elastic properties (in our case, *P*-wave velocities). The main new feature is the application of Monte Carlo inversion to a full 3-D earth model, consisting of Voronoi cells of constant velocity. Both the number of nuclei in the elastic model and the level of the noise in the data are free parameters in the inversion. The inversion is cast in a Bayesian framework and the Monte Carlo algorithm returns a large sample of the parameters of interest that is distributed as in the PPD. The Bayesian formulation makes the sampling results intrinsically parsimonious: the number of Voronoi cells in the sampled velocity models is only as large as necessary to fit the arrival time data. Monte Carlo algorithms are computationally intensive because they require a large number of forward modelling calculations (in this case, ray tracing through the 3-D velocity model). We address this issue by parallelization, launching multiple sampling chains on a multiprocessor computer cluster. This strategy keeps the computation times feasible even for a 3-D inversion (15-30 days total run time). Moreover, by sampling the starting models at random from the prior distribution, we reduce the dependence of the results on the initial point.

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The algorithm is applied to a synthetic data set and to field measurements. The results of the synthetic data inversion demonstrate that the algorithm is able to image velocity anomalies, recover the coordinates of seismic events in space and time, and estimate the level of noise that was added to the data. Our Monte Carlo algorithm applied to field measurements gives results close to those obtained independently by linearized inversion. Monte Carlo sampling enables the quantification of posterior uncertainties, and these uncertainties allow us to visualize regions of the 3-D earth model where the velocities are well resolved by the data.

Further investigations will include a number of upgrades to improve the tomographic inversion and reduce the long run-time. In particular, the introduction of *S*-wave pickings can strongly facilitate event location and can be used to recover the V_P/V_S ratio in the investigated volume. Moreover, the long run-time needed by each inversion can be reduced by adopting a different strategy for the computation of the predicted *P*-wave arrival times, where the geometry of ray paths is not updated at each move (Bodin *et al.* 2012). An additional strategy to reduce run-time and improve mixing in the sampling chains is provided by a 'parallel tempering' approach, where chains interact exchanging information during the MCMC sampling. In this way, the chains converge faster to global minimum and the length of the burn- in phase can be reduced (Dettmer & Dosso 2012; Dosso *et al.* 2012; Sambridge 2014). Run-time can also be decreased in principle by devising a proposal distribution that approximates the posterior and consequently increases the probability of accepting candidate models (e.g. Malinverno 2002). This strategy could save computing time as long as the calculation of the approximate posterior distribution can be done efficiently.

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APPENDIX: EFFECT OF EVENT PRIOR PROBABILITY DISTRIBUTION

In our study, we used prior information about *P*-wave velocities and event locations given in previous studies (Valoroso *et al.* 2009). Here, we illustrate the dependence of the results on the prior probability distribution of the event location in the inversion of field data. In particular, we adopted a prior standard deviation for the event location of 3 km, which is two times the value used in the main text. This high standard deviation allows the events to move substantially in the investigated volume, even assuming unrealistic positions, given the high density of the seismic stations. For this test, all the parameters other than the prior standard deviation of the event locations are the same as the inversion in Section 4.

In Fig. A1(a), we show the mean posterior value for the *P*-wave velocity at 3.0 km depth. For a better comparison between the results of the new inversion and the results presented in Section 4, we used the same colour scale and isolines as in Fig. 10(a) and we plotted the relative difference between the two mean posterior models in Fig. A1(b). Although the structures in the V_P model are very similar between the two inversions (as it can be seen by the density of the isolines in the central, coloured area), the larger prior standard deviation for the event locations decreased the size of such area where the posterior standard deviation of the V_P is smaller than the prior (the brightly coloured area in Fig. A1a). In the outer region of the investigated volume, the effect of larger uncertainties on event locations combines with the small number of ray path crossing such region, to increase the non-uniqueness of the problem, allowing the sampling algorithm to accept a larger number of plausible models.

Fig. A2 displays the spatial density of the event locations along the profile AB in Fig. A1 (compare to Fig. 12b). As in Fig. 12, only events located within 10 km of the vertical section are considered. The larger prior standard deviation of the event locations increased the size of the three clusters highlighted in Fig. 12(b).

In Fig. A3, we plot the posterior mean and standard deviation for position of all the events (compare to Fig. 11). The posterior standard deviations are about twice the values in the inversion in Section 4. However, the vertical locations of the hypocentres are still shifted with respect to the prior mean value, as resulted in the inversion in Section 4.

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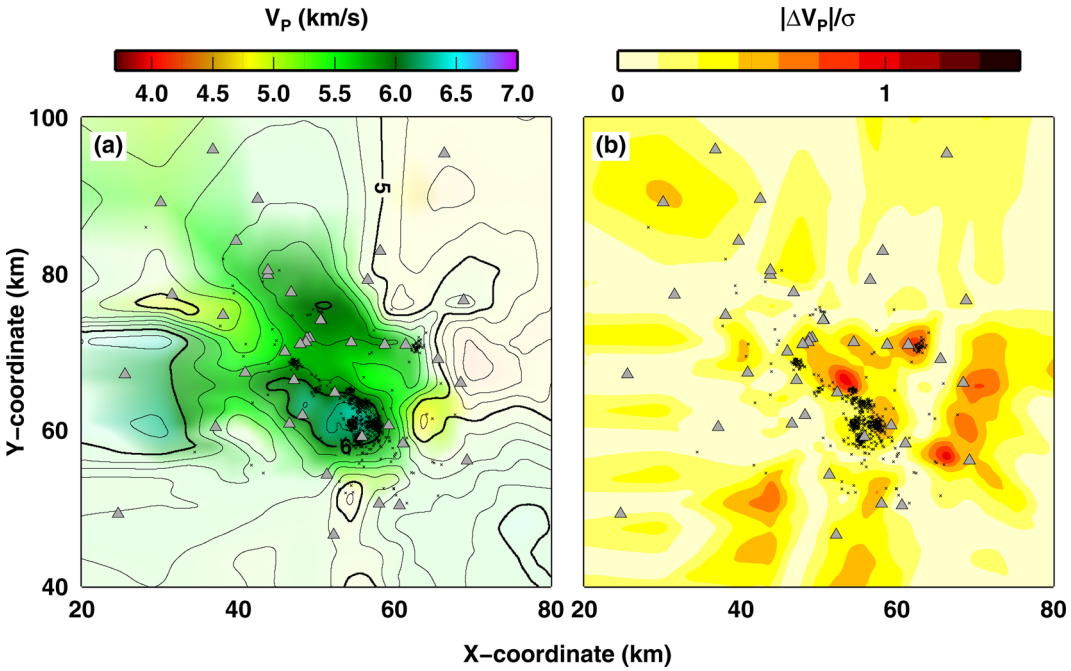


Figure A1. Inversion of field measurements. (a) *P*-wave velocity map at 3.0 km depth, for the inversion using a prior standard deviation for the event position equals to 3.0 km. Symbols as in Fig. 10. Difference between *P*-velocity map in (a) and the one presented in Fig. 10(a). Absolute velocity differences are divided by the posterior standard deviation of *P*-wave velocity from RJMCMC sampling, showing that most differences are within one posterior standard deviation.

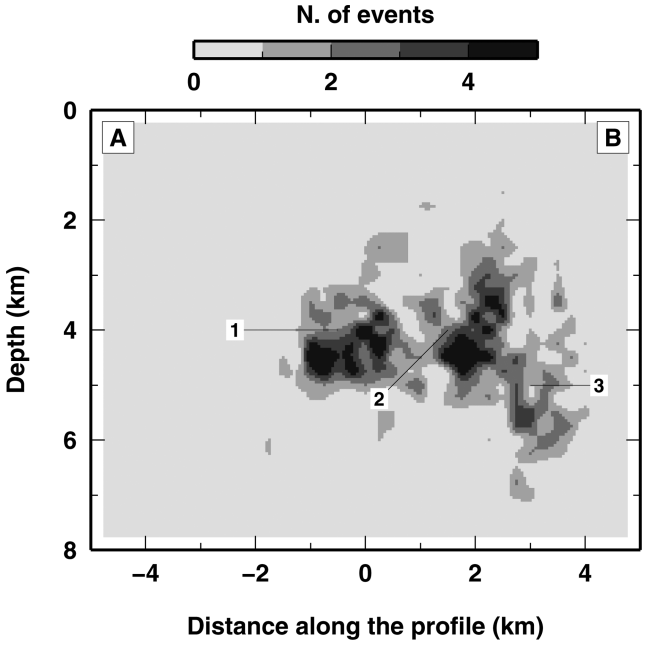


Figure A2. Inversion of field measurements. Event density along profile AB. Inversion using prior standard deviation for the event position equals to 3.0 km. Symbols as in Fig. 12.

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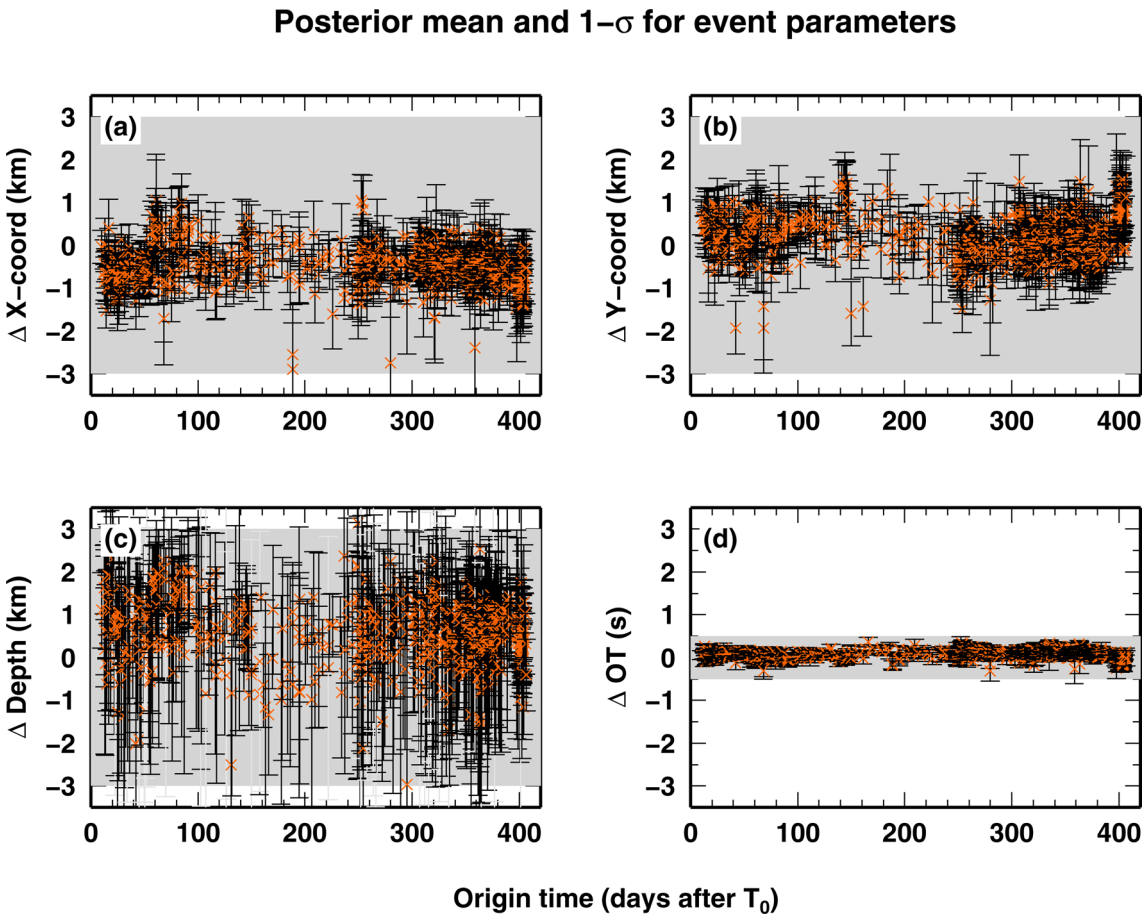


Figure A3. Posterior mean and standard deviation for each event in the inversion of field measurements. Inversion using prior standard deviation for the event position equals to 3.0 km. Symbols as in Fig. 11.

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